

Aaron Weymouth ; Classical Mechanics ; Due 9.10.26

Pset: ch 1: $\{(\check{1}, \check{2}, \check{7}, \check{11}, 18, 20, 22)\}$

Starts PDF: Pg 38/29 book

① Show for Single Particle w/ constant mass that the equation of motion implies: $\frac{dT}{dt} = \vec{F} \cdot \vec{v}$ while mass

varies w/ time corresponding equation

Start w/ $\vec{F} = \frac{d\vec{p}}{dt} = m \frac{d\vec{v}}{dt}$

$\rightarrow$ dot w/ $\vec{v}$ $\vec{F} \cdot \vec{v} = m \vec{v} \cdot \frac{d\vec{v}}{dt}$

Since $\frac{d}{dt}(v^2) = 2\vec{v} \cdot \frac{d\vec{v}}{dt}$

We have

$\vec{F} \cdot \vec{v} = \frac{m}{2} \frac{d(v^2)}{dt} \xrightarrow{\text{for constant } m} \frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = \frac{dT}{dt}$

$$\therefore \frac{dT}{dt} = \vec{F} \cdot \vec{v}$$

for variable mass: $\vec{F} = \frac{d\vec{p}}{dt}$

$\rightarrow$ dot w/ $\vec{p}$: $\vec{F} \cdot \vec{p} = \vec{p} \cdot \frac{d\vec{p}}{dt} = \frac{1}{2} \frac{d(p^2)}{dt}$

but $p^2 = m^2 v^2$

so: $\frac{1}{2} p^2 = m \left(\frac{1}{2} m v^2 \right) = mT$

$$\text{thus: } \frac{d(mT)}{dt} = \vec{F} \cdot \vec{p}$$

here both use $\vec{A} \cdot \vec{A} = \frac{1}{2} \frac{d}{dt} A^2$; for fixed mass $\vec{v}$
for changing mass $\vec{p}$

2. Prove mag R of position vector for COM of an arbitrary origin given by eq:

$$M^2 R^2 = M \sum_i m_i r_i^2 - \frac{1}{2} \sum_{i,j} m_i m_j r_{ij}^2$$

$$\vec{r}_{ij} = \vec{r}_i - \vec{r}_j \quad \therefore \vec{r}_{ij}^2 = (\vec{r}_i - \vec{r}_j)^2 = r_i^2 + r_j^2 - 2\vec{r}_i \cdot \vec{r}_j$$

$$\frac{1}{2} \sum_{i,j} m_i m_j r_{ij}^2 = \frac{1}{2} \sum_{i,j} m_i m_j (r_i^2 + r_j^2 - 2\vec{r}_i \cdot \vec{r}_j)$$

$$= \frac{1}{2} \sum_{i,j} m_i m_j r_i^2 + \frac{1}{2} \sum_{i,j} m_i m_j r_j^2 - \sum_{i,j} m_i m_j \vec{r}_i \cdot \vec{r}_j$$

and $\sum_j m_j = M$

$$\hookrightarrow \frac{M}{2} \sum_i m_i r_i^2 + \frac{M}{2} \sum_j m_j r_j^2 = M \sum_i m_i r_i^2 \quad \text{1st + 2nd term}$$

$$\hookrightarrow \sum_{i,j} m_i m_j \vec{r}_i \cdot \vec{r}_j = \left(\sum_i m_i \vec{r}_i \right) \left(\sum_j m_j \vec{r}_j \right) \quad \text{3rd mix}$$

$$\left[\sum_i m_i m_j \vec{r}_i \cdot \vec{r}_j = MR \right] \hookrightarrow = M^2 R^2$$

$$\text{thus} \rightarrow \frac{1}{2} \sum_{i,j} m_i m_j r_{ij}^2 = M \sum_i m_i r_i^2 - M^2 R^2$$

rearrange:

$$M^2 R^2 = M \sum_i m_i r_i^2 - \frac{1}{2} \sum_{i,j} m_i m_j r_{ij}^2$$

Concept: $\sum_i m_i r_i^2$ describes particle position relative to chosen origin

and r_{ij} is internal separation (like in class)

★ Showing up again: whole system motion
+
internal motion.

7. Show Lagrange equations in form of Eq 1.53

$$\text{Can also be written } \frac{\partial T}{\partial \dot{q}_j} - 2 \frac{\partial T}{\partial q_j} = Q_j$$

Where eq 1.53 is

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = Q_j$$

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = Q_j$$

(1) Total derivative of KE

$$\dot{T} = \sum_k \frac{\partial T}{\partial q_k} \dot{q}_k + \sum_k \frac{\partial T}{\partial \dot{q}_k} \ddot{q}_k + \frac{\partial T}{\partial t}$$

(2) $\rightarrow \partial / \partial \dot{q}_j$

$$\frac{\partial \dot{T}}{\partial \dot{q}_j} = \frac{\partial T}{\partial q_j} + \sum_k \frac{\partial^2 T}{\partial q_k \partial \dot{q}_j} \dot{q}_k + \sum_k \frac{\partial T}{\partial \dot{q}_k \partial \dot{q}_j} \ddot{q}_k + \frac{\partial^2 T}{\partial t \partial \dot{q}_j}$$

here the last three terms are exactly $\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right)$

$$\hookrightarrow \frac{\partial \dot{T}}{\partial \dot{q}_j} = \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) + \frac{\partial T}{\partial q_j} \quad \text{w/ right side} = Q_j$$

$$\therefore \frac{\partial \dot{T}}{\partial \dot{q}_j} - 2 \frac{\partial T}{\partial q_j} = Q_j$$

This form isn't a new equation. Just Lagrange written

differently. The extra coordinate derivative appears

bc T contains $\partial T / \partial q_k \cdot \dot{q}_k$ so when I differentiate

$\dot{q}_k$ w/ $\dot{q}_k$ I get j^{th} term

⑪ Consider a uniform thin disk that rolls w/o slipping on same horizontal plane.

We have horizontal force applied on center of disk and in a direction parallel to plane.

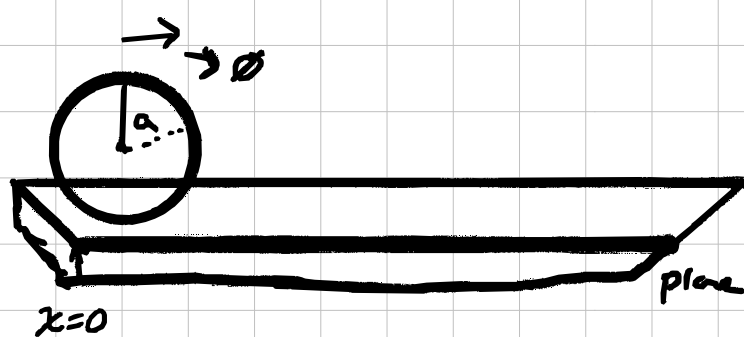
(a) derive lagrange + find generalized

(b) discuss motions of force not applied in parallel.

(a):

$N=1$ disk

has both rot + translational motion.



x is $\rightarrow$ horizontal displacement

θ is rotation angle of disk rotation. Coordinates are related by rolling constraint.

$$x = a\theta, \quad \dot{x} = a\dot{\theta} \quad \star 1 \text{ independent generalized coordinate}$$

$$q = x.$$

The KE contains both translation and rotation

$$T = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} I \dot{\theta}^2 \quad (\text{here } \dot{\theta} = \dot{x}/a)$$

$$T = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} I \left(\frac{\dot{x}^2}{a^2} \right) = \frac{1}{2} \left(m + \frac{I}{a^2} \right) \dot{x}^2$$

The center remains at constant height ($\Delta y = 0$)

So V is constant $\rightarrow$ may be taken as zero.

$$L = T - V = \frac{1}{2} \left(m + \frac{I}{a^2} \right) \dot{x}^2 \quad \text{then the generalized force: } Q_x = \vec{F} \cdot \frac{\partial \vec{r}_c}{\partial x}$$

Since

$$\vec{r}_c = x\hat{x} + a\hat{y}, \quad \frac{\partial \vec{r}_c}{\partial x} = \hat{x}; \quad \vec{F} = F\hat{x}$$

$$Q_x = F; \quad \text{there is one generalized coordinate} \\ \therefore \text{one lagrangian equation}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) - \frac{\partial L}{\partial x} = Q_x$$

$$\text{i) } \frac{\partial L}{\partial x} = \frac{\partial}{\partial x} \left(\left(\frac{1}{2} m + \frac{I}{a^2} \right) \dot{x}^2 \right) = 0$$

$$\text{ii) } \frac{\partial L}{\partial \dot{x}} = \frac{\partial}{\partial \dot{x}} \left(\left(\frac{1}{2} m + \frac{I}{a^2} \right) \dot{x}^2 \right) = \left(m + \frac{I}{a^2} \right) \dot{x}$$

$$\therefore \left(m + \frac{I}{a^2} \right) \ddot{x} = F \quad \text{for a uniform thin disc.}$$

$$I = \frac{1}{2} m a^2 \quad \text{so} \quad \frac{3}{2} m \ddot{x} = F$$

$$\therefore \ddot{x} = \frac{2F}{3m}, \quad \ddot{\theta} = \frac{2F}{3ma}$$

The rolling constraint couples translation and rotation in terms of x , the rotational kinetic energy

appears as an additional effective inertia I/a^2

So disk behaves as though

$$m_{\text{eff}} = m + \frac{I}{a^2}$$

$\rightarrow$ constraint force (friction) need not appear in the lagrangian explicitly bc No-slip constraint has already be incorporated thru $x = a\theta$

b. the horizontal force is not parallel to disc plane, $\rightarrow$ resolve it to:

$$F_{\parallel} = F \cos \theta; \quad F_{\perp} = F \sin \theta$$

The parallel component produces the rolling motion described. Perpendicular attempts to move disc sideways and that can't be described by a single constraint ($x = a\theta$)

$\therefore$ One coordinate description is not sufficient

if static friction prevents lateral slipping

additional generalized coordinates + constraints are required to describe disc orientation and motion.

if required static cannot be supplied

disk slips sideways.

1.11 When 2 pool balls collide instantaneous forces are large

but only for infinitesimal time Δt such that $\int_{\Delta t} \vec{F} \cdot d\vec{v}$ is finite.

these are impulsive forces. Show that if impulses are present Lagrange equations may be transformed into:

$$\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right)_f - \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right)_i = S_j$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = Q_j$$

During the collision, Q_j contains the impulsive force

Integrating thru short collision @ interval $t_i \rightarrow t_f$

$$\int_{t_i}^{t_f} \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) dt - \int_{t_i}^{t_f} \frac{\partial \mathcal{L}}{\partial q_j} dt = \int_{t_i}^{t_f} Q_j dt$$

↓

$$\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right)_f - \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right)_i$$

Book defines generalized impulse

$$S_j \equiv \int_{t_i}^{t_f} Q_j dt \quad \dots \quad \text{Since collision time is infinitesimal while non-impulsive term remains finite:}$$

$$\int_{t_i}^{t_f} \frac{\partial \mathcal{L}}{\partial q_j} dt \rightarrow 0$$

and therefore:

$$\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right)_f - \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right)_i = S_j$$

$$\text{or since } p_j = \frac{\partial \mathcal{L}}{\partial \dot{q}_j} \rightarrow p_{j,f} - p_{j,i} = S_j$$

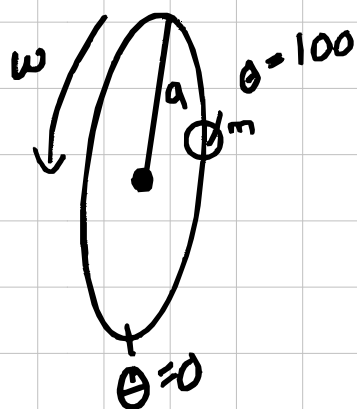
★ Over an infinitesimal collision time, the integral of any finite non-impulsive term vanishes, while impulsive force has a finite integral.

$$Q_j = Q_j^{(\text{impulsive})} + Q_j^{(\text{non-impulsive})}$$

and only the impulsive contribution survives the integration.

1.18 a point particle on massless hoop

- fixed radius a ; rotates about vertical symmetry axis
 - constant angular speed ω
 - Obtain Lagrangian w/ assumption only gravity is external
 - Show $\omega > \omega_0 \rightarrow$ particle in stationary loop; and
if $\omega < \omega_0$; particle only at bottom;
 - what is ω_0 ?
-



Let θ be particle's angle measured from bottom of hoop.

$$\rho = a \sin \theta; z = -a \cos \theta; \phi = \omega t$$

- Since our hoop rotates rigidly about the vert:

$$\dot{\phi} = \omega$$

$\rightarrow$ Speed of bead has 2 pieces:

$$v^2 = a^2 \dot{\theta}^2 + a^2 \omega^2 \sin^2 \theta$$

$$KE = T = \frac{1}{2} m a^2 (\dot{\theta}^2 + \omega^2 \sin^2 \theta)$$

problem constraints V to only gravity:

$$V = mgz = -mga \cos(\theta)$$

$$L = T - V = \frac{1}{2} m a^2 \dot{\theta}^2 + \frac{1}{2} m a^2 \omega^2 \sin^2 \theta + m g a \cos \theta$$

1.18 cont:

$$\text{for } \theta: \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) - \frac{\partial \mathcal{L}}{\partial \theta} = 0$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} &= \frac{\partial}{\partial \dot{\theta}} \left(\frac{1}{2} m a^2 \dot{\theta}^2 + \frac{1}{2} m a^2 \omega^2 \sin^2 \theta + m g a \cos \theta \right) \\ &= m a^2 \dot{\theta} \quad (i) \end{aligned}$$

$$\text{then } \frac{d}{dt} (m a^2 \dot{\theta}) = m a^2 \ddot{\theta} \quad (ii)$$

$$\begin{aligned} \text{iii) } \frac{\partial \mathcal{L}}{\partial \theta} &= \frac{\partial}{\partial \theta} \left(\frac{1}{2} m a^2 \dot{\theta}^2 + \frac{1}{2} m a^2 \omega^2 \sin^2 \theta + m g a \cos \theta \right) \\ &= m a^2 \omega^2 \sin \theta \cos \theta - m g a \sin \theta \end{aligned}$$

thus:

$$m a^2 \ddot{\theta} - m a^2 \omega^2 \sin \theta \cos \theta + m g a \sin \theta = 0$$

↳ Equivalently:

$$\ddot{\theta} = \sin \theta \left(\omega^2 \cos \theta - \frac{g}{a} \right)$$

I was unsure which Pr 1B was intended; I have 2x version of book: doing other 1.1B pr here.

→ give $\mathcal{L}$ and asks for EOM $\rightarrow \mathcal{L}' = \frac{m}{2}(a\dot{x}^2 + 2b\dot{x}\dot{y} + c\dot{y}^2) - \frac{k}{2}(a\dot{x}^2 + 2b\dot{x}\dot{y} + c\dot{y}^2)$

$$\text{Let } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad q = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\text{then } \mathcal{L}' = \frac{m}{2} \dot{q}^T A \cdot \dot{q} - \frac{k}{2} q^T A q \quad \text{so the E-L is}$$

$$A(m\ddot{q} + kq) = 0$$

$$\text{Since } \det(A) = ac - b^2 \neq 0$$

$$m\ddot{q} + kq = 0$$

$$\hookrightarrow \ddot{x} + \frac{k}{m}x = 0; \quad \ddot{y} + \frac{k}{m}y = 0$$

Clearly system is 2d-isotropic h.o.

$$\hookrightarrow \text{for } a=c=0$$

$$\mathcal{L}' = mb\dot{x}\dot{y} - kbxy$$

but variation w/ respect to x and y yields the oscillatory

eq for y & x .

$$\text{for } b=0, \quad c=-a$$

$$\mathcal{L}' = \frac{a}{2} [m(\dot{x}^2 - \dot{y}^2) - k(x^2 - y^2)]$$

★ NO real point transformation makes KE positive def.

P. 1.20: 1d particle w/ given Lagrangian; V is differentiable function of x . Find EOM ($x(t)$) + describe.

given: $\mathcal{L} = \frac{m^2 \dot{x}^4}{12} + m \dot{x}^2 V(x) - V(x)^2$

→ $V(x)$ does have units of Energy, $[L] = \text{energy}^2$
 ↳ Not normal E-L

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{m^2}{3} \dot{x}^3 + 2mV\dot{x} \quad \dots (1)$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = m^2 \dot{x}^2 \ddot{x} + 2mV\ddot{x} + 2mV'\dot{x}^2 \quad \dots (2)$$

$$\frac{\partial \mathcal{L}}{\partial x} = m\dot{x}^2 V' - 2VV' \quad \dots (3)$$

E-L:

$$\underbrace{m^2 \dot{x}^2 \ddot{x} + 2mV\ddot{x}} + \underbrace{2mV'\dot{x}^2 - m\dot{x}^2 V' + 2VV'} = 0$$

$$(m^2 \dot{x}^2 + 2mV)\ddot{x} + (m\dot{x}^2 + 2V)V' = 0$$

$$(m\dot{x}^2 + 2V)(m\ddot{x} + V') = 0$$

↳ from right: $\boxed{m\ddot{x} = -V'(x)}$

Clearly particle of mass m moving in conservative potential

velocity: $\frac{\partial^2 \mathcal{L}}{\partial \dot{x}^2} = m^2 \dot{x}^2 + 2mV = m(m\dot{x}^2 + 2V)$

↳ $m\dot{x}^2 + 2V = 0$ [Lagrangian becomes singular here]

Let Mechanical Energy:

$$E_0 = \frac{1}{2} m \dot{x}^2 + V$$

then $m\dot{x}^2 + 2V = 2E_0$

↳ system is at $E_0 = 0$

Use turning points w/ derivative

$$\dot{E}_0 = \dot{x}(m\ddot{x} + V') = 0$$

↳ so for $\dot{x} \neq 0 \rightarrow$ we get Newton.

at $E_0 = 0$; E-L loses its rank

so division by that factor should be stated as assumption.

Lastly $\mathcal{L}$ has no time dependence explicitly:

$$\mathcal{E}_L = \dot{x} \frac{\partial \mathcal{L}}{\partial \dot{x}} - \mathcal{L} \quad \rightarrow \text{sub in}$$

$$\mathcal{E}_L = \dot{x} \left(\frac{m^2}{3} \dot{x}^3 + 2mV\dot{x} \right) - \left(\frac{m^2 \dot{x}^4}{12} + mV\dot{x}^2 - V^2 \right)$$

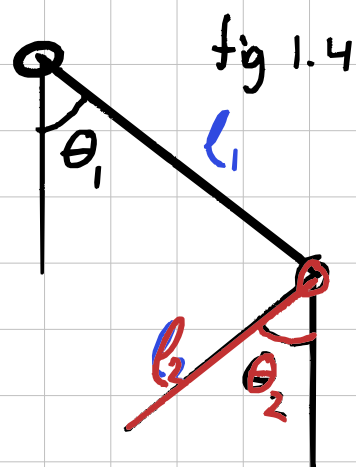
$$= \frac{1}{4} m^2 \dot{x}^4 + mV\dot{x}^2 + V^2$$

$$= \left(\frac{1}{2} m \dot{x}^2 + V \right)^2$$

$$\boxed{\therefore \mathcal{E}_L = E_0^2}$$

★ Lagrange generates same Newton dynamics
 but is conserved energy is square
 of the ME.

P. 1.22 Double Pendulum:



Let $\Delta = \theta_1 - \theta_2$

$$(m_1) \quad x_1 = l_1 \sin \theta_1, \quad y_1 = -l_1 \cos \theta_1$$

$$(m_2) \quad x_2 = l_1 \sin \theta_1 + l_2 \sin \theta_2 \quad y_2 = -l_1 \cos \theta_1 - l_2 \cos \theta_2$$

KE: $v_1^2 = \dot{x}_1^2 + \dot{y}_1^2 = l_1^2 \dot{\theta}_1^2$

$$\begin{aligned} v_2^2 &= (l_1 \dot{\theta}_1 \cos \theta_1 + l_2 \dot{\theta}_2 \cos \theta_2)^2 + (l_1 \dot{\theta}_1 \sin \theta_1 + l_2 \dot{\theta}_2 \sin \theta_2)^2 \\ &= l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \end{aligned}$$

$$\therefore T = \frac{1}{2} (m_1 + m_2) l_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 l_2^2 \dot{\theta}_2^2 + m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2)$$

↳ cross term is coupling between m_1 and m_2 .

PE: $V = m_1 g y_1 + m_2 g y_2$

$$V = -(m_1 + m_2) g l_1 \cos \theta_1 - m_2 g l_2 \cos \theta_2$$

$$\begin{aligned} \mathcal{L} &= \frac{1}{2} (m_1 + m_2) l_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 l_2^2 \dot{\theta}_2^2 + m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\Delta) \\ &\quad + (m_1 + m_2) g l_1 \cos \theta_1 + m_2 g l_2 \cos \theta_2 \end{aligned} \quad \dots (1)$$

$$\text{Eq for } \theta_1: \frac{\partial \mathcal{L}}{\partial \theta_1} = (m_1 + m_2) l_1^2 \ddot{\theta}_1 + m_2 l_1 l_2 \ddot{\theta}_2 \cos \Delta$$

$$\hookrightarrow \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_1} \right) = (m_1 + m_2) l_1^2 \ddot{\theta}_1 + m_2 l_1 l_2 [\ddot{\theta}_2 \cos \Delta - \dot{\theta}_2 (\dot{\theta}_1 - \dot{\theta}_2) \sin \Delta]$$

$$\frac{\partial \mathcal{L}}{\partial \theta_1} = -m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin \Delta - (m_1 + m_2) g l_1 \sin \theta_1$$

use sympy python (provided)

$$(m_1 + m_2) l_1^2 \ddot{\theta}_1 + m_2 l_1 l_2 \cos(\Delta) \ddot{\theta}_2 + m_2 l_1 l_2 \sin(\Delta) \dot{\theta}_1^2 + (m_1 + m_2) g l_1 \sin \theta_1 = 0$$

$$\text{Equation for } \theta_2: \frac{\partial \mathcal{L}}{\partial \dot{\theta}_2} = m_2 l_2^2 \ddot{\theta}_2 + m_2 l_1 l_2 \dot{\theta}_1 \cos(\Delta)$$

$$\frac{\partial \mathcal{L}}{\partial \theta_2} = m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\Delta) - m_2 g l_2 \sin \theta_2$$

$$\therefore \text{E-L: } m_2 l_2^2 \ddot{\theta}_2 + m_2 l_1 l_2 \cos(\Delta) \ddot{\theta}_1 - m_2 l_1 l_2 \sin(\Delta) \dot{\theta}_1^2 + m_2 g l_2 \sin(\theta_2) = 0$$

quick checks on D.P.:

- Each term has units of Energy ✓
- for small angles: $\sin \theta_i \approx \theta_i$, $\cos(\Delta) \approx 1$, $\dot{\theta}_i^2 \sin(\Delta) = O(\theta^3)$

$$\hookrightarrow (m_1 + m_2) l_1^2 \ddot{\theta}_1 + m_2 l_1 l_2 \ddot{\theta}_2 + (m_1 + m_2) g l_1 \theta_1 = 0 \quad \checkmark$$

$$\hookrightarrow m_2 l_2^2 \ddot{\theta}_2 + m_2 l_1 l_2 \ddot{\theta}_1 + m_2 g l_2 \theta_2 = 0$$

Linear coupled oscillator.

$\mathcal{L}$ is time independent; total ME. conserved (Lecture 9.8.26)